Uncertainty relations from majorization and entropy in the binary case

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Majorization uncertainty relations use a single state-independent probability vector to bound a whole family of Schur-concave uncertainty measures. We study two-outcome measurements and determine when their optimal bounds are attained. For noncommuting projective qubit measurements, neither the optimal direct-sum nor tensor-product vector is realized by any state, so every induced bound for a continuous strictly Schur-concave measure is strict; the tensor-product Rényi bounds are tight only at orders zero and infinity. For projective qubits, we also quantify the gap between universal and entropy-specific bounds. We derive the optimal direct-sum vector for arbitrary noisy and biased two-outcome qubit POVMs and characterize its attainability completely. Unbiased unsharpness preserves ideal-detector uncertainty relations, but bias can invalidate them and can produce spurious steering violations when ideal bounds are used. We give a necessary and sufficient condition for these relations to survive and illustrate the corrected steering criterion through a closed-form Werner-state analysis. We also extend both bounds to binary projective measurements in arbitrary finite dimension and the direct-sum bound to binary POVMs.

I. INTRODUCTION

The uncertainty principle is one of the defining features of quantum mechanics [1–3]: there are pairs of measurements whose outcomes cannot both be predicted with certainty, however carefully the system is prepared. This preparation uncertainty is distinct from measurement uncertainty, which limits how accurately incompatible observables can be measured jointly [4]; we are concerned with the former. First formulated for the position and momentum of a particle, the principle has since become a quantitative tool of quantum information science. It underlies the security of quantum key distribution [5–7], provides experimentally accessible criteria for entanglement [6, 8, 9] and for Einstein–Podolsky–Rosen steering [10–12], and is closely related to, though distinct from, the incompatibility of quantum measurements [13, 14]. These applications call for formulations that apply to measurements with a finite number of outcomes and whose bounds do not depend on the state of the system.

Robertson's variance-based relation [2] does not meet these requirements: its lower bound depends on the state, it can vanish even for incompatible observables, and variances are not natural when the outcomes are mere labels [15]. Entropies of the outcome distributions provide a remedy [15–19]. The best known entropic uncertainty relation, due to Maassen and Uffink [20], bounds the sum of the Shannon entropies of the outcomes of two measurements by a quantity that depends only on the largest overlap between their eigenvectors. Entropic uncertainty relations have since grown into a broad field with applications throughout quantum information theory [21, 22]; they have been refined in various regimes [23, 24] and tested in experiments [25, 26].

Each such relation, however, constrains a single measure of uncertainty, while different tasks call for different measures: the Shannon entropy quantifies average information, the min-entropy governs guessing probabilities [27], the Rényi family interpolates between these and other extremes, and the Tsallis family generalizes the Shannon entropy in a different way. Majorization [28], the order that captures when one probability distribution is more disordered than another, makes it possible to treat all of them at once. It is a central tool of quantum information theory, where it governs, for instance, the transformations of entangled states [29] and, more generally, quantum resource theories [30, 31]. In the context of uncertainty, the idea is to find a single probability vector, independent of the state, that majorizes the outcome statistics of every state; every uncertainty measure that respects majorization then obeys a bound determined by this one vector. Depending on whether the two outcome distributions are combined by a direct sum or by a tensor product, such measures include the Shannon and Tsallis entropies or the Shannon and all Rényi entropies, among many others. This program was initiated by Partovi [32] and developed into universal uncertainty relations by Friedland, Gheorghiu and Gour [33] and by Puchała, Rudnicki and Życzkowski [34]; Rudnicki *et al.* subsequently introduced a direct-sum formulation that yields stronger entropic bounds [35]. Majorization uncertainty relations have since been extended to generalized measurements, mixed states and quantum operations [36–38], sharpened further [39–42], in particular by exploiting the lattice structure of majorization, veri-

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fied experimentally [42, 43], and turned into steering witnesses [44].

The price of this universality is only partly understood. For a qubit measured in two bases—the setting of many experiments and, for mutually unbiased bases, of the BB84 protocol [5]—entropy-specific relations can be optimized: the optimal Shannon relation is known [45, 46], in closed form over part of the range and through a one-parameter minimization otherwise, and so are optimal relations for Rényi entropies [47, 48]. Earlier majorization bounds have been compared with the Maassen–Uffink relation and with the optimal Shannon bound [37, 49], and the optimal direct-sum bound for two qubit bases has been identified in the majorization lattice [40]. Partovi already remarked that the optimal majorization vector is in general not the statistics of any state, so that a minimum-uncertainty state need not exist in this framework [32], a point recently reiterated in Ref. [50]. Li and Qiao also observed that the partial sums of this bound are usually reached by different states and that, in examples, its Shannon entropy falls short of the optimal Shannon bound [51]. What is missing is a complete account of when such bounds are tight, why, and how this depends on the uncertainty measure. In the simplest setting, where both approaches can be carried out completely, what exactly does majorization gain, and what does it cost?

In this paper we answer this question for pairs of binary (two-outcome) measurements, and the answer is simple to state. Unless the measurements commute, no state attains the optimal majorization bound, so every bound it implies for a strictly Schur-concave measure of uncertainty, such as the Shannon entropy, is strict; for the tensor-product Rényi bounds, only orders zero and infinity are tight. The price is substantial: for two qubit bases the direct-sum Shannon bound falls short of the exact minimum by up to 22%. Even so, it outperforms the Maassen–Uffink relation unless the measurement axes on the Bloch sphere are within about nine degrees of orthogonal, i.e., unless the measurements are nearly mutually unbiased [37], and its ratio to the exact minimum tends to one, though only logarithmically, as the measurements approach compatibility, where the Maassen–Uffink relation is weakest. The optimal bound for noisy and biased measurements also has a practical consequence: detector noise can only increase uncertainty, but detector bias, produced for instance by mismatched efficiencies or click/no-click detection, can push the statistics outside the ideal quantum set. A single inequality decides when relations derived for ideal detectors survive; when it fails, an uncertainty-based steering test can certify steering for a product state. For the Shannon entropy alone, a known closed-form bound [45, 49] is stronger than both at every noncommuting angle; the case for majorization rests on its universality, not on its sharpness for any single measure.

Several ingredients of our analysis are known. For two qubit bases, the direct-sum relation is a special case of the construction of Rudnicki *et al.* [35], and its optimality and saturating states were obtained by Li and Qiao [40]. The tensor-product vector follows from the general construction of Friedland, Gheorghiu and Gour [33], also for binary projective measurements in higher dimensions; for qubits it also coincides with the vector $Q^{(1)}$ of Puchała *et al.* [34]. The ellipse of achievable qubit expectation values [49, 52, 53], the eigenvalue characterization of direct-sum partial sums [37], and the optimality of the flattened direct-sum vector [42] are likewise known.

Our contributions are fourfold. First, we give a complete attainment analysis for two-outcome qubit measurements. For noncommuting projective measurements, we prove that neither optimal vector is realized by any state, making rigorous the general remark of Partovi [32] and the observations of Ref. [51]. This implies strictness for every continuous strictly Schur-concave measure. For arbitrary two-outcome qubit POVMs, we combine the noncommuting obstruction with an explicit necessary and sufficient condition for attainment in the commuting case.

Second, we evaluate the optimal direct-sum vector for arbitrary noisy and biased two-outcome qubit POVMs in closed form and clarify its relation to joint measurability. Third, we determine exactly when the ideal-detector relation survives imperfect detection: for qubits this reduces to $\Lambda \le 2\sqrt{c}$. We show how detector bias can invalidate ideal steering tests and provide a corrected criterion, illustrated by a closed-form Werner-state analysis with efficiency threshold $\eta_* = (4+\sqrt{2})/7$.

Fourth, we specialize the general constructions to binary measurements in arbitrary finite dimension, show that the resulting vectors are already ordered, and determine attainment for binary projective measurements. Thus, the higher-dimensional tensor-product vector itself is not claimed as new; the new elements are its explicit binary evaluation, the observation that no flattening is required, and the attainment analysis. We also quantify the gap with Maassen–Uffink and entropy-specific optima and compare the bounds across the Rényi family, distinguishing analytical results from numerical observations throughout.

The paper is organized as follows. Section II introduces majorization, the uncertainty measures and the measurements we consider, and Sec. III describes the geometry of the achievable statistics. Section IV derives the optimal bounds for projective qubit measurements, and Sec. V discusses their entropic consequences and compares them with entropy-specific relations. Section VI treats general two-outcome qubit measurements, which may be noisy or biased and are described by positive operator-valued measures (POVMs), i.e., sets of positive operators that sum to the identity, and the consequences of detector imperfections; Sec. VII treats binary measurements in arbitrary dimension. Section VIII discusses the results, including their relation to quantum optics, and concludes. All analytical results were also checked numerically (Appendix A).

II. PRELIMINARIES

A. Majorization and uncertainty measures

For $x \in \mathbb{R}^n$ let $x^{\downarrow}$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^{k} x_i^{\downarrow}$ its Lorenz partial sums, with $L_0(x) = 0$; the Lorenz curve of x is the piecewise-linear interpolation of the points $(k, L_k(x))$, $k = 0, \ldots, n$. For x, y with equal total, x is *majorized* by y, written $x \prec y$, if $L_k(x) \le L_k(y)$ for all k [28]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \ge \Phi(y)$, and *strictly* Schur-concave if, in addition, $\Phi(x) > \Phi(y)$ whenever $x \prec y$ and $x^{\downarrow} \neq y^{\downarrow}$. Every function $\Phi(x) = \sum_i \phi(x_i)$ with ϕ (strictly) concave is (strictly) Schur-concave [28].

The uncertainty measures used below are the Rényi [54] and Tsallis [55] entropies

$$H_\alpha(p) = \frac{\log_2 \sum_i p_i^\alpha}{1-\alpha}, \qquad T_\gamma(p) = \frac{1 - \sum_i p_i^\gamma}{\gamma - 1}, \tag{1}$$

with $\alpha \in [0, \infty]$ and $\gamma > 0$; the cases $\alpha \in \{0, 1, \infty\}$ and $\gamma = 1$ are defined by continuity, with the Shannon entropy $H = H_1$ as the common limit ($T_\gamma \to H \ln 2$ as $\gamma \to 1$). The orders $\alpha = \infty$ and $\alpha = \frac{1}{2}$ give the min-entropy $H_\infty(p) = -\log_2 \max_i p_i$ and the max-entropy $H_{1/2}$ [27], and H_0 is the logarithm of the number of non-zero entries. All of them are Schur-concave; H_α is strictly Schur-concave for $0 < \alpha < \infty$ and T_γ for all $\gamma > 0$, whereas H_0 and H_∞ are not.

Let $p(\rho)$ and $q(\rho)$ denote the outcome distributions of two measurements M and N on a state ρ; we also write $H_\alpha(M) = H_\alpha(p)$ and $H_\alpha(N) = H_\alpha(q)$. For measurements in orthonormal bases $\{|m_i\rangle\}$ and $\{|n_j\rangle\}$, the Maassen–Uffink (MU) relation [20] reads

$$H(p) + H(q) \;\ge\; -\log_2 c, \qquad c = \max_{i,j} |\langle m_i | n_j \rangle|^2. \tag{2}$$

It constrains one particular functional of the pair (p, q); a different uncertainty measure requires a different relation. A majorization uncertainty relation [32–35] instead consists of a single state-independent vector ω such that

$$p(\rho) \oplus q(\rho) \;\prec\; \omega \quad \text{or} \quad p(\rho) \otimes q(\rho) \;\prec\; \omega \qquad \forall \rho. \tag{3}$$

We call these the direct-sum and the tensor-product form, respectively. Every Schur-concave function Φ then inherits the bound $\Phi(p \oplus q) \ge \Phi(\omega)$ or $\Phi(p \otimes q) \ge \Phi(\omega)$. Sums of Shannon or Tsallis entropies are Schur-concave functions of $p \oplus q$, since, the totals of p and q being fixed, they are sums of a concave function over all entries; sums of Shannon or Rényi entropies are Schur-concave functions of $p \otimes q$, since $H_\alpha(p \otimes q) = H_\alpha(p) + H_\alpha(q)$. The direct-sum form also yields bounds for Rényi entropies of order $\alpha \le 1$, by a separate argument (Corollary 1), but not for all orders (Proposition 2).

Restricted to non-negative, non-increasing vectors of fixed length and fixed total, $\prec$ is a partial order, and in fact a complete lattice [56, 57].

Definition 1. Let $\mathcal{A}$ be a set of non-negative vectors of common length and total. A vector ω is an *optimal majorization bound* for $\mathcal{A}$ if $a \prec \omega$ for all $a \in \mathcal{A}$, and $\omega \prec \omega'$ for every ω' with this property; i.e., $\omega = \sup \mathcal{A}$ in the majorization lattice.

Lemma 1 ([56, 57])**.** *Let $S_k = \sup_{a \in \mathcal{A}} L_k(a)$. Then $\sup \mathcal{A}$ is the vector whose Lorenz curve is the least concave majorant of $k \mapsto S_k$ (with $S_0 = 0$). In particular, if $\omega_k = S_k - S_{k-1}$ is already non-increasing, then $\omega = \sup \mathcal{A}$.*

In all cases treated below the second, simpler statement applies. The quantities S_k have a transparent meaning: S_k is the largest value that any state can give to a sum of k entries, i.e., to the summed probabilities of k outcomes.

B. Two-outcome qubit measurements

A qubit state is $\rho = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma})$ with $|\vec{r}| \le 1$. A projective two-outcome measurement is $M = \{P_\pm\}$ with $P_\pm = \frac{1}{2}(\mathbb{1} \pm \hat{m} \cdot \vec{\sigma}) = |m_\pm\rangle\langle m_\pm|$, $|\hat{m}| = 1$, and similarly $N = \{Q_\pm\}$ with axis $\hat{n}$ and eigenvectors $|n_\pm\rangle$. The outcome distributions are

$$p = \left(\tfrac{1+x}{2}, \tfrac{1-x}{2}\right), \qquad q = \left(\tfrac{1+y}{2}, \tfrac{1-y}{2}\right), \tag{4}$$

where $x = \hat{m} \cdot \vec{r}$ and $y = \hat{n} \cdot \vec{r}$. The overlaps are $|\langle m_\pm | n_\pm \rangle|^2 = \frac{1}{2}(1 + \hat{m} \cdot \hat{n})$ and $|\langle m_\pm | n_\mp \rangle|^2 = \frac{1}{2}(1 - \hat{m} \cdot \hat{n})$, hence

$$c = \tfrac{1}{2}\big(1 + |\hat{m} \cdot \hat{n}|\big) = \cos^2 \tfrac{\theta}{2}, \qquad \cos\theta = |\hat{m} \cdot \hat{n}|, \tag{5}$$

where $\theta \in [0, \frac{\pi}{2}]$ is the (acute) Bloch angle between the two measurement axes. Relabeling the outcomes of N maps $\hat{n} \mapsto -\hat{n}$ and does not affect any majorization statement, so we may and shall assume $\hat{m} \cdot \hat{n} = \cos\theta \ge 0$. Note that $\frac{1}{2} \le c \le 1$ and $\sqrt{c} \ge 1/\sqrt{2} > 1 - \sqrt{c}$.

A general two-outcome POVM $\{E_0, E_1 = \mathbb{1} - E_0\}$ can always be written as

$$E_0 = \tfrac{1}{2}\big[(1+\mu)\mathbb{1} + \eta\, \hat{a} \cdot \vec{\sigma}\big], \qquad \eta \ge 0,\ |\mu| + \eta \le 1, \tag{6}$$

with $|\hat{a}| = 1$, the constraint being equivalent to $0 \le E_0 \le \mathbb{1}$. Here η is the sharpness and μ the bias; rank-one projective measurements correspond to $\eta = 1$, $\mu = 0$.

III. GEOMETRY OF THE ACHIEVABLE SET

The statistics (4) depend on $\vec{r}$ only through (x, y).

Lemma 2 ([49, 52, 53])**.** *For non-parallel axes, the set $\{(\hat{m} \cdot \vec{r}, \hat{n} \cdot \vec{r}) : |\vec{r}| \le 1\}$ is the filled ellipse*

$$\mathcal{E}_\theta = \big\{(x, y) : x^2 - 2xy\cos\theta + y^2 \le \sin^2\theta\big\}. \tag{7}$$

Its boundary consists exactly of the images of the pure states whose Bloch vectors lie in $\mathrm{span}\{\hat{m}, \hat{n}\}$*. Moreover, for all $s, t \in \mathbb{R}$,*

$$\max_{|\vec{r}| \le 1} (sx + ty) = |s\hat{m} + t\hat{n}|. \tag{8}$$

Proof. Write $\vec{r} = \vec{r}_\parallel + \vec{r}_\perp$ with $\vec{r}_\parallel \in \mathrm{span}\{\hat{m}, \hat{n}\}$. Only $\vec{r}_\parallel$ contributes to (x, y), and $(x, y) = G\xi$ where ξ are the coefficients of $\vec{r}_\parallel$ in the basis $\{\hat{m}, \hat{n}\}$ and $G = \left(\begin{smallmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{smallmatrix}\right)$ is the Gram matrix. Then $|\vec{r}_\parallel|^2 = \xi^T G \xi = (x, y) G^{-1} (x, y)^T \le 1$, which is the stated inequality since $\det G = \sin^2\theta$; equality holds iff $\vec{r}_\perp = 0$ and $|\vec{r}| = 1$. Conversely, every $(x, y) \in \mathcal{E}_\theta$ is attained by $\vec{r} = \xi_1 \hat{m} + \xi_2 \hat{n}$ with $\xi = G^{-1}(x, y)^T$, since then $|\vec{r}|^2 = (x, y) G^{-1} (x, y)^T \le 1$. Eq. (8) is Cauchy–Schwarz: $sx + ty = \vec{r} \cdot (s\hat{m} + t\hat{n})$. □

Equivalently, $(x, y) \in \mathcal{E}_\theta$ if and only if the covariance matrix of the observables $A = \hat{m} \cdot \vec{\sigma}$ and $B = \hat{n} \cdot \vec{\sigma}$,

$$\Gamma = \begin{pmatrix} 1 - x^2 & \cos\theta - xy \\ \cos\theta - xy & 1 - y^2 \end{pmatrix}, \tag{9}$$

is positive semidefinite, because $\det \Gamma = \sin^2\theta - (x^2 - 2xy\cos\theta + y^2)$. Its off-diagonal entry contains the anticommutator $\frac{1}{2}\{A, B\} = \cos\theta\, \mathbb{1}$, which does not depend on the state. The ellipse is a disk for mutually unbiased measurements ($\theta = \pi/2$) and degenerates into the segment $x = y$ as $\theta \to 0$ [53].

Consequently the achievable set $\mathcal{P}_\oplus(M, N) = \{p(\rho) \oplus q(\rho)\} \subset \mathbb{R}^4$ is the image of $\mathcal{E}_\theta$ under the affine map $(x, y) \mapsto \frac{1}{2}(1 + x, 1 - x, 1 + y, 1 - y)$: it is convex and compact, and its extreme points are the statistics of pure states in the plane of the two axes (Fig. 1). Since $\mathcal{E}_\theta$ is centered at the origin, moving a point radially outward to $\partial\mathcal{E}_\theta$ multiplies $|x|$ and $|y|$ by the same factor, which makes both p and q more ordered. Every uncertainty measure of the form $\Phi(p) + \Psi(q)$ with Φ and Ψ Schur-concave, in particular every Rényi and Tsallis sum, is therefore minimized on $\partial\mathcal{E}_\theta$, and the optimization problems below are effectively one-dimensional.

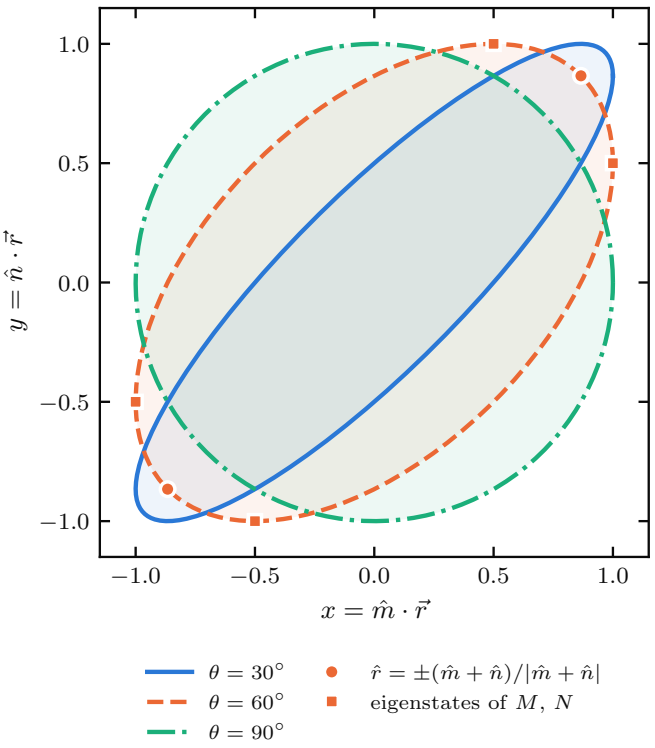


FIG. 1. The ellipse $\mathcal{E}_\theta$ of achievable expectation values $(x, y) = (\langle \hat{m} \cdot \vec{\sigma} \rangle, \langle \hat{n} \cdot \vec{\sigma} \rangle)$ for three Bloch angles. Markers (for $\theta = 60°$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r} \propto \pm(\hat{m} + \hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

IV. PROJECTIVE MEASUREMENTS

A. Direct-sum bound

Theorem 1 ([35, 40])**.** *Let M, N be projective qubit measurements with maximal overlap c. For every state ρ,*

$$p(\rho) \oplus q(\rho) \prec \omega_\oplus := \big(1,\ \sqrt{c},\ 1 - \sqrt{c},\ 0\big), \tag{10}$$

and $\omega_\oplus$ is the optimal majorization bound for $\mathcal{P}_\oplus(M, N)$.

The majorization relation (10) is the qubit case of the direct-sum relation $p \oplus q \prec \{1\} \oplus W$ of Rudnicki *et al.* [35]: for $d = 2$ their vector $W = (s_1, s_2 - s_1)$ has $s_1 = \sqrt{c}$ and $s_2 = 1$. Its optimality, and the states that saturate its partial sums, were obtained by Li and Qiao in the majorization lattice [40]. We give a short proof based on Lemma 2.

Proof. We compute $S_k = \max_\rho L_k(p \oplus q)$ for $k = 1, \dots, 4$. Clearly $S_1 \le 1$ and $S_4 = 2$. Since all entries are nonnegative, $S_3 = 2 - \min_\rho \min_i (p \oplus q)_i \le 2$. Both bounds are attained by an eigenstate of M, for which $p = (1, 0)$. For $k = 2$, the two largest entries either belong to the same distribution, with sum 1, or are of the form $p_i + q_j$. By Lemma 2,

$$p_i + q_j = 1 + \tfrac{1}{2}(\pm x \pm y) \le 1 + \tfrac{1}{2} \max\{|\hat{m} + \hat{n}|, |\hat{m} - \hat{n}|\}. \tag{11}$$

With $\hat{m} \cdot \hat{n} = \cos\theta \ge 0$ the maximum is $|\hat{m} + \hat{n}| = \sqrt{2(1 + \cos\theta)} = 2\cos\frac{\theta}{2} = 2\sqrt{c}$, so

$$S_2 = 1 + \sqrt{c}, \tag{12}$$

attained by the bisector state $\vec{r} = (\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$. This is the qubit case of the Landau–Pollak-type inequality $p_i + q_j \le 1 + \sqrt{c}$ underlying Deutsch's relation [15, 58]. The increments of $(S_1, S_2, S_3, S_4) = (1, 1 + \sqrt{c}, 2, 2)$ give $\omega_\oplus$, which is non-increasing because $1 \ge \sqrt{c} \ge 1 - \sqrt{c} \ge 0$. Lemma 1 completes the proof. □

B. Tensor-product bound

Theorem 2 ([32–34])**.** *Under the assumptions of Theorem 1, for every state ρ,*

$$p(\rho) \otimes q(\rho) \prec \omega_\otimes := \big(w,\ 1 - w,\ 0,\ 0\big), \tag{13}$$

where $w = \frac{1}{4}\big(1+\sqrt{c}\big)^2$*, and* $\omega_\otimes$ *is the optimal majorization bound for* $\mathcal{P}_\otimes(M,N) = \{p(\rho)\otimes q(\rho)\}$.

The vector $\omega_\otimes$ coincides with the vector $Q^{(1)} = (C^2, 1-C^2, 0, 0)$, $C = (1+\sqrt{c})/2$, of Puchała *et al.* [34], which for $d = 2$ exhausts their hierarchy of bounds. It is also the vector of Friedland *et al.* [33], whose entries are the increments of the maximal partial sums S_k; for qubits these increments are non-increasing, so that $\omega_\otimes$ is the least upper bound. For mutually unbiased bases it is the vector found by Partovi [32].

Proof. By the arithmetic–geometric mean inequality and Eq. (12), $p_i q_j \le \big(\frac{p_i+q_j}{2}\big)^2 \le \big(\frac{1+\sqrt{c}}{2}\big)^2 = w$, with equality for the bisector state, for which $p_+ = q_+ = (1+\sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M, $p\otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \ge 1/\sqrt{2}$ we have $w \ge \frac{1}{2}$, so the increments $(w, 1-w, 0, 0)$ are non-increasing, and Lemma 1 applies. □

C. Attainability

Neither optimal bound is the statistics of a state unless the measurements commute.

Proposition 1. *(i) There is a state with* $(p\oplus q)^\downarrow = \omega_\oplus$ *if and only if* $c = 1$*, i.e., M and N commute. For* $c < 1$*, the maxima* S_1, S_3 *are attained only by the eigenstates of M and N, and* S_2 *only by the bisector states* $\vec{r} = \pm(\hat{m}+\hat{n})/|\hat{m}+\hat{n}|$ *(for* $\theta = \pi/2$ *also by* $\vec{r} = \pm(\hat{m}-\hat{n})/|\hat{m}-\hat{n}|$*). (ii) There is a state with* $(p\otimes q)^\downarrow = \omega_\otimes$ *if and only if* $c = 1$.

Proof. (i) $(p\oplus q)^\downarrow_1 = 1$ forces ρ to be an eigenstate of M or N, say $\vec{r} = \pm\hat{m}$. Then $q = \big(\frac{1\pm\cos\theta}{2}, \frac{1\mp\cos\theta}{2}\big)$ and $(p\oplus q)^\downarrow = (1, c, 1-c, 0)$, which equals $\omega_\oplus$ iff $c = \sqrt{c}$, i.e., $c = 1$. The statement about S_2 follows from the equality condition in Cauchy–Schwarz and from $|\hat{m}-\hat{n}| < |\hat{m}+\hat{n}|$ for $\theta < \pi/2$. (ii) If $c = 1$, a common eigenstate of M and N gives $p\otimes q = (1,0,0,0) = \omega_\otimes$. If $c < 1$, $L_2(p\otimes q) = 1 = L_2(\omega_\otimes)$ requires two vanishing entries of $p\otimes q$, hence an eigenstate of M or N, at which $L_1(p\otimes q) = c < w = L_1(\omega_\otimes)$, since $w - c = \frac{1}{4}(1-\sqrt{c})(1+3\sqrt{c}) > 0$. □

Proposition 1 makes precise the sense in which $\omega_\oplus$ and $\omega_\otimes$ are *descriptions* of the achievable sets rather than the statistics of a particular state: each is a supremum in the majorization lattice, and the partial sums of $\omega_\oplus$ are saturated by two disjoint families of states (Fig. 2). Demanding certainty about M lowers the largest attainable $L_2(p\oplus q)$ from $1+\sqrt{c}$ to $1+c$; this deficit, $\sqrt{c}(1-\sqrt{c})$, vanishes only for commuting measurements. Proposition 1 makes precise, for binary measurements, the general remark of Partovi that the optimal majorization vector need not be the statistics of any state [32]. Once the states that saturate the individual partial sums are known [40], its proof is short; its value lies in the consequences derived below.

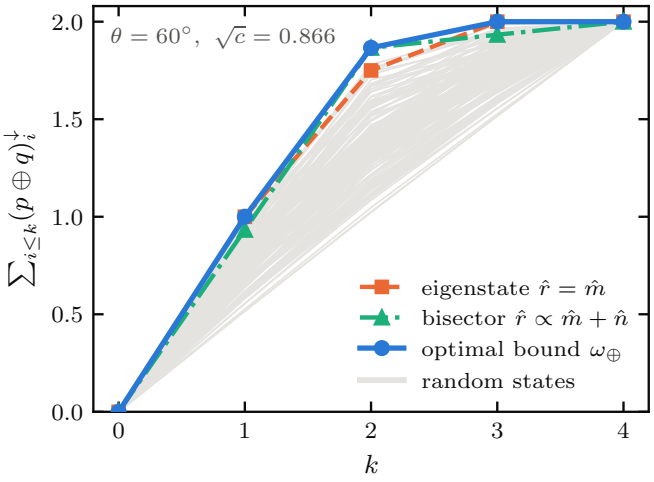


FIG. 2. Lorenz curves of $p\oplus q$ for $\theta = 60°$. Grey: 150 random states. The optimal bound $\omega_\oplus$ (blue) is the smallest concave curve above all of them; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it. A similar plot for $\theta = 90°$ is Fig. 1(a) of Ref. [40].

V. ENTROPIC UNCERTAINTY RELATIONS

Let $h(t) = -t\log_2 t - (1-t)\log_2(1-t)$ be the binary entropy.

A. Bounds for the Rényi and Tsallis families

Corollary 1 ([35])**.** *For every concave* ϕ *on* $[0,1]$, $\sum_i \phi(p_i) + \sum_j \phi(q_j) \ge \phi(1) + \phi(\sqrt{c}) + \phi(1-\sqrt{c}) + \phi(0)$. *In particular,*

$$H(p) + H(q) \;\ge\; h(\sqrt{c}), \tag{14}$$

$$T_\gamma(p) + T_\gamma(q) \;\ge\; \frac{1 - c^{\gamma/2} - (1-\sqrt{c})^\gamma}{\gamma - 1} \qquad (\gamma > 0), \tag{15}$$

and, for $0 < \alpha \le 1$,

$$H_\alpha(p) + H_\alpha(q) \;\ge\; H_\alpha(\sqrt{c}, 1-\sqrt{c}), \tag{16}$$

where $H_\alpha(\sqrt{c}, 1-\sqrt{c}) = \log_2\big[c^{\alpha/2} + (1-\sqrt{c})^\alpha\big]/(1-\alpha)$.

Proof. The first statement is Karamata's inequality applied to Theorem 1. Eqs. (14) and (15) follow with $\phi(t) = -t\log_2 t$ and $\phi_\gamma(t) = (t - t^\gamma)/(\gamma-1)$, since $\sum_i p_i = \sum_j q_j = 1$ gives $T_\gamma(p) + T_\gamma(q) = \sum_k \phi_\gamma\big((p\oplus q)_k\big)$. Eq. (16) is the qubit case of Theorem 2 of Ref. [35]; we recall the short argument. For $\alpha < 1$ write $\sum_i p_i^\alpha = 1 + X$ and $\sum_j q_j^\alpha = 1 + Y$ with $X, Y \ge 0$. Then

$$\begin{aligned} H_\alpha(p) + H_\alpha(q) &= \frac{\log_2[(1+X)(1+Y)]}{1-\alpha} \\ &\ge \frac{\log_2(1+X+Y)}{1-\alpha}, \end{aligned}$$

and $1 + X + Y = \sum_k (p \oplus q)_k^\alpha - 1 \geq \sum_k (\omega_\oplus)_k^\alpha - 1 = c^{\alpha/2} + (1-\sqrt{c})^\alpha$ by Schur-concavity of $v \mapsto \sum_k v_k^\alpha$. □

Corollary 2 ([33, 34])**.** *For every $\alpha \in [0, \infty]$,*

$$H_\alpha(p) + H_\alpha(q) \;\geq\; H_\alpha(\omega_\otimes) = \frac{\log_2\left[w^\alpha + (1-w)^\alpha\right]}{1-\alpha}. \tag{17}$$

For $\alpha = 1$ this reads $H(p) + H(q) \geq h(w)$; for $\alpha = \infty$ it gives $H_\infty(p) + H_\infty(q) \geq -2\log_2 \frac{1+\sqrt{c}}{2}$, which is tight (bisector state); since $H \geq H_\infty$, it contains Deutsch's Shannon bound [15], which has the same right-hand side.

As noted in Sec. II, the two forms suit different families of entropies. Where both apply, the direct sum is stronger, but it cannot be extended to all orders. That the direct-sum Rényi relation does not hold for $\alpha > 1$ was stated in Ref. [35] (Sec. III); part (ii) below gives an explicit counterexample:

Proposition 2. *(i) For $0 < \alpha \leq 1$, $H_\alpha(\sqrt{c}, 1-\sqrt{c}) \geq H_\alpha(\omega_\otimes)$, with equality iff $c = 1$; in particular $h(\sqrt{c}) \geq h(w)$. (ii) For $c < 1$ the direct-sum form (16) fails at $\alpha = \infty$: $\min_\rho [H_\infty(p) + H_\infty(q)] = -2\log_2 \frac{1+\sqrt{c}}{2} < -\frac{1}{2}\log_2 c = H_\infty(\sqrt{c}, 1-\sqrt{c})$.*

Proof. (i) Since $w - \sqrt{c} = \frac{1}{4}(1-\sqrt{c})^2 \geq 0$, we have $(\sqrt{c}, 1-\sqrt{c}) \prec (w, 1-w)$, with equality iff $c = 1$; apply strict Schur-concavity. The analogous statement in any dimension is proven in Ref. [35]. (ii) With $t = \sqrt{c}$, $\left(\frac{1+t}{2}\right)^2 - t = \left(\frac{1-t}{2}\right)^2 > 0$ for $t < 1$; the minimum is the tight bound of Corollary 2. □

The failure, however, sets in only beyond $\alpha = 2$. Numerically, for qubits $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ is a valid bound for every $\alpha \leq 2$ at every angle, and it fails above an angle-dependent order between 2.00 ($\theta \to 0$) and 2.58 ($\theta \to 90°$). At $\alpha = 2$ the bisector state is optimal at every angle [48], and with $s = \sqrt{c}$ and $a = \frac{1+s}{2}$ the exact minimum is $-2\log_2[a^2 + (1-a)^2] = -2\log_2 \frac{1+s^2}{2}$, whereas $H_2(s, 1-s) = -\log_2[s^2 + (1-s)^2]$; the bound holds iff $\left(\frac{1+s^2}{2}\right)^2 \leq s^2 + (1-s)^2$, i.e., iff $\frac{1}{4}(s-1)^3(s+3) \leq 0$, which is true for $s \leq 1$. The direct-sum form thus extends well beyond the range $\alpha \leq 1$ in which it is proven in general.

B. When are the bounds tight?

Theorem 3. *Let $c < 1$. (i) For every continuous, strictly Schur-concave Φ,*

$$\min_\rho \Phi(p \oplus q) > \Phi(\omega_\oplus), \qquad \min_\rho \Phi(p \otimes q) > \Phi(\omega_\otimes). \tag{18}$$

Consequently the bounds (14), (15) and (16), and the bounds (17) with $0 < \alpha < \infty$, are all strict. (ii) The bounds (17) of order $\alpha = 0$ and $\alpha = \infty$ are tight; they are attained by the eigenstates of M or N and by the bisector states, respectively.

Proof. (i) The achievable statistics form a compact set, so the minimum of Φ is attained at some state ρ^*. By Theorem 1 and Proposition 1(i), $p \oplus q(\rho^*) \prec \omega_\oplus$ with $(p \oplus q)^\downarrow \neq \omega_\oplus$, hence $\Phi(p \oplus q(\rho^*)) > \Phi(\omega_\oplus)$. The tensor product is analogous, by Theorem 2 and Proposition 1(ii). The Shannon and Tsallis sums are strictly Schur-concave functions of $p \oplus q$, and $H_\alpha(p) + H_\alpha(q) = H_\alpha(p \otimes q)$ with H_α strictly Schur-concave for $0 < \alpha < \infty$. For (16) with $\alpha < 1$, apply the argument to $v \mapsto \sum_k v_k^\alpha$ in the proof of Corollary 1. (ii) An eigenstate of M gives $H_0(p) + H_0(q) = 0 + 1 = H_0(\omega_\otimes)$, and a bisector state gives $\max_i p_i \max_j q_j = w$. □

The proof of (i) uses only that the achievable statistics form a compact set whose least upper bound is not attained. It therefore applies whenever the optimal bound is not attained: to every pair of qubit POVMs whose effects do not commute (Proposition 3) and to both bounds for binary projective measurements in any dimension with $c < 1$ (Theorem 5).

Theorem 3 makes precise in what sense the majorization bounds are optimal. As majorization bounds they cannot be improved (Theorems 1 and 2); yet, because the supremum is not the statistics of any state, the bound it induces is strict for every strictly Schur-concave measure. For such measures the optimum has to be found measure by measure, which is what entropy-specific relations do; for the tensor-product Rényi bounds, only at the extreme orders $\alpha = 0$ and $\alpha = \infty$ is the majorization bound tight.

C. Comparison with the Maassen–Uffink relation and with the exact optimum

Figure 3 compares the Shannon bounds with MU and with the exact minimum of $H(p) + H(q)$, which is known for qubits [45, 46]. By Sec. III the minimum lies on the great circle through $\hat{m}$ and $\hat{n}$. For $\theta \leq \bar{\theta}$ it is attained at the bisector states and equals $2h\left(\frac{1+\sqrt{\bar{c}}}{2}\right)$ [46]. At $\bar{\theta}$ the bisector turns from a local minimum into a local maximum along the circle; a second-derivative test shows that this happens when $\sqrt{\bar{c}}\,\operatorname{artanh}\sqrt{\bar{c}} = 1$ [59], i.e., at $\bar{c} \approx 0.6948$ and $\bar{\theta} \approx 67.07°$. For $\bar{\theta} < \theta < \pi/2$ the minimizers bifurcate and the minimum has to be computed numerically [46]. As θ grows they move continuously towards the eigenstates of M and N, which they reach at $\theta = \pi/2$, where MU is tight.

The range in which the bisector is optimal depends on the uncertainty measure, as Zozor *et al.* showed for Rényi entropies [48]; that the relative uncertainty of qubit states depends on the measure reflects a lack of majorization between their statistics [60]. For $c > \frac{1}{2}$ the eigenstates are optimal for $\alpha \leq \alpha_0(c)$, the bisector for $\alpha \geq \alpha^\star(c)$, and intermediate states in between. Here α_0 exceeds $\frac{1}{2}$ (about 0.52, 0.56 and 0.66 at 20°, 60° and 85°), and $\alpha^\star$ decreases from $\alpha^\dagger \approx 1.43$ as $c \to \frac{1}{2}$ to $\frac{1}{2}$ as $c \to 1$; for mutually unbiased measurements the eigenstates remain optimal for all $\alpha < \alpha^\dagger$, and the bisector

takes over at $\alpha^\dagger$. In terms of the Bloch angle, the bisector is optimal up to 43° for $\alpha = 3/4$, up to $\bar\theta \approx 67°$ for the Shannon entropy and up to 80° for $\alpha = 1.2$, and at all angles for $\alpha \geq \alpha^\dagger$, in line with the tightness of the min-entropy bound. Here $\alpha^\dagger$ solves $2H_\alpha\big(\frac{2+\sqrt{2}}{4}, \frac{2-\sqrt{2}}{4}\big) = 1$, the condition for bisector and eigenstates to perform equally well when the measurements are mutually unbiased. For the max-entropy, for instance, the eigenstates give $\log_2(1+\sin\theta)$ and the bisector $2\log_2(1+\sin\frac{\theta}{2})$.

For mutually unbiased measurements ($\theta = 90°$, $c = \frac{1}{2}$) MU is tight (1 bit), while $h(1/\sqrt{2}) \approx 0.872$ and $h(w) \approx 0.844$. As the measurements become more compatible, however, the majorization bounds take over: $h(\sqrt{c}) > -\log_2 c$ for $\sqrt{c} > t^* \approx 0.7585$, i.e., for Bloch angles $\theta < \theta^* \approx 81.3°$ (the tensor bound overtakes MU at $\theta < 79.3°$), as reported numerically in Ref. [37]; earlier majorization bounds [33, 34] were compared with MU and with the optimal qubit bound in Ref. [49]. For the Shannon entropy alone, however, neither majorization bound is the best closed-form relation available. The bound

$$H(p) + H(q) \;\geq\; h\Big(\frac{1+\sqrt{2c-1}}{2}\Big) \qquad (19)$$

of Refs. [45, 49] (see also Ref. [22]), where $2c-1 = \cos\theta$, is strictly stronger than $h(\sqrt{c})$ for every $0 < \theta \leq \pi/2$ and at least as strong as MU, with which it coincides, and hence is tight, at $\theta = 90°$ (Fig. 3); at $\theta = 60°$, for instance, it gives 0.601 bit against 0.568 for $h(\sqrt{c})$. What majorization offers instead is a single bound for all Schur-concave measures at once.

As $c \to 1$ the advantage over MU becomes unbounded, and the direct-sum bound becomes tight in ratio. With $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$ one has $-\log_2 c \approx 2\varepsilon/\ln 2$ and $h(\sqrt{c}) = h(\varepsilon) \approx \varepsilon\log_2(e/\varepsilon)$, whereas the exact minimum is $2h(\varepsilon/2) \approx \varepsilon\log_2(2e/\varepsilon)$. To leading order the ratio of the exact minimum to $h(\sqrt{c})$ is therefore $1 + 1/\log_2(e/\varepsilon)$: it tends to 1, but only logarithmically (1.09 at $\theta = 0.1$ rad, 1.047 at $\theta = 0.003$ rad), while the ratio of $h(\sqrt{c})$ to MU grows as $\frac{1}{2}\ln(8e/\theta^2)$ (3.8 and 7.3). The eigenstates of M, although they saturate S_1 and S_3, give $h(c)$, asymptotically twice the minimum (the ratio approaches 2 slowly; it is 1.67 and 1.82 at the same angles).

Between these limits the strictness of Theorem 3 is far from marginal. Over all angles the direct-sum bound $h(\sqrt{c})$ reaches at least 78% of the exact minimum, the smallest ratio, 0.782, occurring at $\theta \approx 70°$, just beyond $\bar\theta$, where the minimizer leaves the bisector; the absolute gap is largest, 0.196 bit, at $\theta \approx 74°$, and the tensor-product bound $h(w)$ reaches at least 76.1% of the exact minimum (at $\theta \approx 70.6°$) and falls short by up to 0.217 bit. At $\theta = 90°$, where MU is tight, $h(\sqrt{c})$ still misses the exact value 1 bit by 0.128 bit. Wherever a single strictly Schur-concave measure is what matters, such as the Shannon entropy in the analysis of key distribution, this gap is the price of universality, and the entropy-specific optimum should be used.

Figure 4 extends the comparison to the whole Rényi

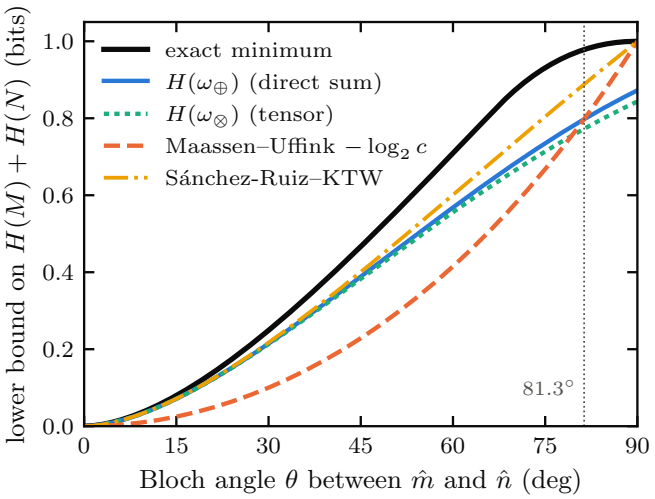


FIG. 3. Lower bounds on $H(M) + H(N)$ as a function of the Bloch angle θ ($c = \cos^2\frac{\theta}{2}$). Black: exact minimum, equal to $2h\big(\frac{1+\sqrt{c}}{2}\big)$ for $\theta \leq \bar\theta \approx 67.07°$. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen–Uffink. Yellow (dash-dotted): the closed-form Shannon bound (19) of Refs. [45, 49], which is strictly stronger than $h(\sqrt{c})$ at every noncommuting angle and at least as strong as MU. The direct-sum bound is stronger than MU for $\theta < 81.3°$ (dotted vertical line).

family at $\theta = 60°$ and $\theta = 85°$. For $\alpha \leq 1$ MU also applies, because H_α is non-increasing in α. At $\theta = 60°$ both majorization bounds beat MU for all $\alpha \leq 1$; at $\theta = 85°$, closer to mutual unbiasedness, they do so only for $\alpha < 0.69$ (direct sum) and $\alpha < 0.58$ (tensor product). For the max-entropy, $\alpha = \frac{1}{2}$, the direct-sum bound reaches 83% and 91% of the exact minimum at the two angles. Numerically, the expression $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ is observed to remain a valid bound throughout $1 < \alpha \leq 2$ at every sampled angle, but it fails above $\alpha \approx 2.1$ ($\theta = 60°$) and $\alpha \approx 2.5$ ($\theta = 85°$), in line with Proposition 2(ii). For large α only the tensor-product bound remains, and it converges to the exact minimum as $\alpha \to \infty$. The majorization bounds thus become exact only at the two ends of the family, as Theorem 3 requires.

Finally, MU is one member of a family $H_\alpha(p) + H_\beta(q) \geq -\log_2 c$ with $1/\alpha + 1/\beta = 2$ [20], which includes the pair $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$ of min- and max-entropy. Majorization relations of the form (3) are symmetric under the exchange of p and q and therefore yield such mixed-order relations only indirectly, through the monotonicity of H_α in α. Conditional majorization bounds, which constrain q for a given p, yield them directly: for qubits they give the full uncertainty region of any pair (H_α, H_β) [41].

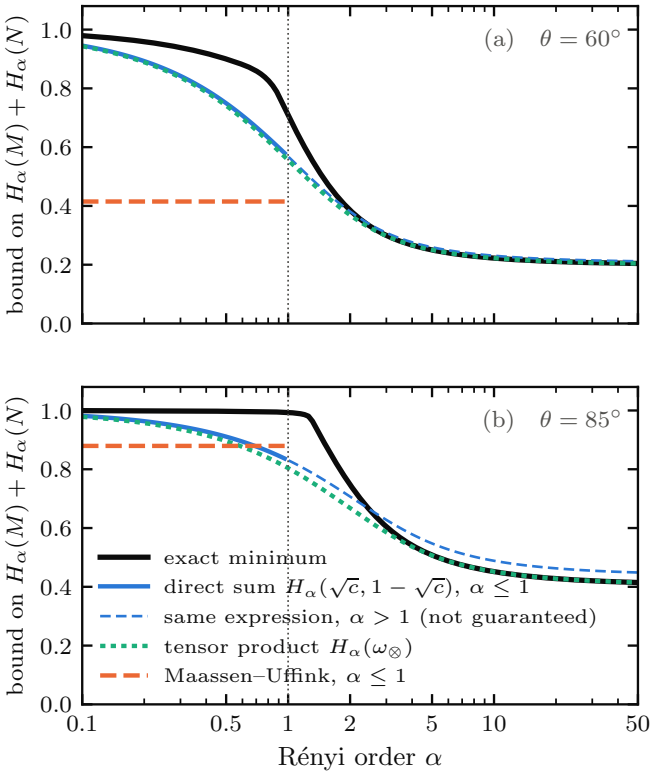


FIG. 4. Bounds on $H_\alpha(M) + H_\alpha(N)$ across the Rényi family for (a) $\theta = 60°$ and (b) $\theta = 85°$. Black: exact minimum. Blue: direct-sum bound $H_\alpha(\sqrt{c}, 1-\sqrt{c})$ of Corollary 1, proven for $\alpha \le 1$; its thin dashed continuation for $\alpha > 1$ is numerically observed to remain a bound up to $\alpha = 2$ but exceeds the exact minimum above $\alpha \approx 2.1$ in (a) and $\alpha \approx 2.5$ in (b). Green (dotted): tensor-product bound $H_\alpha(\omega_\otimes)$ of Corollary 2. Orange (dashed): MU, valid for $\alpha \le 1$. The dotted vertical line marks the Shannon case $\alpha = 1$.

VI. GENERAL TWO-OUTCOME QUBIT POVMS

For POVMs, the partial sums of the direct-sum relation are largest eigenvalues of sums of effects, $S_k = \max_{|J|=k} \lambda_{\max}\big(\sum_{j\in J} K_j\big)$, where the K_j run over the effects of both measurements [37] (see also Refs. [35, 36, 61]); Baek *et al.* evaluated them for unbiased unsharp qubit observables [37], and the flattened vector these partial sums define is optimal for general measurements [42]. For two-outcome qubit POVMs, whose achievable expectation values were described geometrically in Ref. [53], the maximization can be carried out in closed form for arbitrary bias, and the resulting vector turns out to be ordered, so that no flattening is needed.

Let $M = \{E_0, E_1\}$ with parameters $(\mu, \eta, \hat{a})$ and $N = \{F_0, F_1\}$ with $(\nu, \zeta, \hat{b})$, as in Eq. (6). Then $p = \frac{1}{2}\big(1+\mu+\eta\,\hat{a}\cdot\vec{r},\ 1-\mu-\eta\,\hat{a}\cdot\vec{r}\big)$, and similarly for q.

Theorem 4. *Define*

$$\kappa = \max\{|\mu| + \eta,\ |\nu| + \zeta\}, \tag{20}$$

$$\Lambda = \max_{s,t=\pm1} \big[s\mu + t\nu + |s\eta\,\hat{a} + t\zeta\,\hat{b}|\big]. \tag{21}$$

Then $p(\rho) \oplus q(\rho) \prec \omega_\oplus(M,N)$ for all ρ, where

$$\omega_\oplus(M,N) = \tfrac{1}{2}\big(1+\kappa,\ 1+\Lambda-\kappa,\ 1+\kappa-\Lambda,\ 1-\kappa\big) \tag{22}$$

is the optimal majorization bound.

Proof. The largest single outcome probability is $\max_\rho \frac{1}{2}\big(1 + |\mu + \eta\,\hat{a}\cdot\vec{r}|\big) = \frac{1}{2}(1 + |\mu| + \eta)$ for M, so $S_1 = \frac{1}{2}(1+\kappa)$; likewise the smallest is $\frac{1}{2}(1-\kappa)$, so $S_3 = 2 - \frac{1}{2}(1-\kappa)$. For S_2, a pair within one distribution sums to 1, while a cross pair gives, by the Cauchy–Schwarz inequality as in Eq. (8),

$$\max_\rho \big[1 + \tfrac{1}{2}\big(s(\mu + \eta\,\hat{a}\cdot\vec{r}) + t(\nu + \zeta\,\hat{b}\cdot\vec{r})\big)\big] = 1 + \tfrac{1}{2}\big[s\mu + t\nu + |s\eta\hat{a} + t\zeta\hat{b}|\big]. \tag{23}$$

Choosing $s = \operatorname{sgn}\mu$ and $t = \operatorname{sgn}\nu$ (either sign if zero) shows $\Lambda \ge 0$, so $S_2 = 1 + \frac{1}{2}\Lambda$. The increments of (S_k) are the entries of Eq. (22). They are non-increasing iff $\kappa \le \Lambda \le 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda \le |\mu| + |\nu| + \eta + \zeta \le 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M; then $S_2 \ge S_1 + \max_j q_j(\rho^*) \ge S_1 + \frac{1}{2}$, i.e., $\Lambda \ge \kappa$. Lemma 1 completes the proof. □

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$ and Eq. (22) reduces to Eq. (10). For unbiased measurements ($\mu = \nu = 0$), $\Lambda = (\eta^2 + \zeta^2 + 2\eta\zeta|\hat{a}\cdot\hat{b}|)^{1/2}$, in agreement with Ref. [37]. If moreover $\eta = \zeta$, then $\omega_\oplus(M,N) = \eta\,\omega_\oplus + \frac{1-\eta}{2}(1,1,1,1)$, where $\omega_\oplus$ is the vector (10) of the sharp measurements along $\hat{a}$ and $\hat{b}$: such measurements are noisy versions of sharp ones [37, 62], and the achievable statistics, their supremum and the saturating states are those of the sharp case mixed with the uniform vector. For unequal sharpness or nonzero bias, $\omega_\oplus(M,N)$ is not of this form, and the relevant geometric quantities are the lengths $|\eta\hat{a} \pm \zeta\hat{b}|$ rather than the overlap c. Theorem 4 also covers states of fixed spectrum: projective measurements on states with $|\vec{r}| = r$ have the statistics of unbiased measurements of sharpness $\eta = \zeta = r$ on pure states, so the spectrum-dependent bound [38, 40] is $\frac{1}{2}\big(1+r,\ 1-r+2r\sqrt{c},\ 1+r-2r\sqrt{c},\ 1-r\big)$, and by Proposition 3 below it is not attained for $r > 0$ and $c < 1$.

As for projective measurements, the bound is not attained when the effects do not commute; with bias, it may not be attained even when they do.

Proposition 3. *Let M and N be as in Theorem 4. (i) If E_0 and F_0 do not commute, i.e., $\eta\zeta\,\hat{a}\times\hat{b} \ne 0$, no state attains $\omega_\oplus(M,N)$. (ii) If E_0 and F_0 commute, then, after relabeling outcomes and choosing axes for any trivial*

measurements so that $\hat{b} = \hat{a}$*, define*

$$A_\varepsilon = \frac{1 + |\mu + \varepsilon\eta|}{2}, \qquad B_\varepsilon = \frac{1 + |\nu + \varepsilon\zeta|}{2} \qquad (\varepsilon = \pm 1). \tag{24}$$

A state attains $\omega_\oplus(M, N)$ *if and only if some* ε *satisfies both*

$$\max\{A_\varepsilon, B_\varepsilon\} = \max_{\delta=\pm1} \max\{A_\delta, B_\delta\}, \tag{25}$$

$$A_\varepsilon + B_\varepsilon = \max_{\delta=\pm1}(A_\delta + B_\delta). \tag{26}$$

The attaining state is $\vec{r} = \varepsilon\hat{a}$*. (iii) With bias, commuting effects need not attain it: for* $\hat{b} = \hat{a}$*,* $(\mu, \eta) = (0.3, 0.6)$ *and* $(\nu, \zeta) = (-0.8, 0.15)$*,* S_1 *is attained only at* $\vec{r} = -\hat{a}$ *and* S_2 *only at* $\vec{r} = \hat{a}$*.*

Proof. (i) Suppose a state attains $S_1 = \frac{1}{2}(1+\kappa)$ through an outcome of M, so that $\kappa = |\mu| + \eta$; the case of N is symmetric. Since $\eta > 0$, this forces $\vec{r} = s_0\hat{a}$ with $s_0\mu = |\mu|$. There the two largest entries of $p \oplus q$ sum to $1 + \frac{1}{2}\max_t\big[|\mu| + \eta + t\nu + ts_0\zeta\,\hat{a}\cdot\hat{b}\big]$, whereas the term $s = s_0$ of Λ gives $S_2 \geq 1 + \frac{1}{2}\big[|\mu| + t\nu + |\eta\hat{a} + ts_0\zeta\hat{b}|\big]$ for each t. By Cauchy–Schwarz, $|\eta\hat{a} + ts_0\zeta\hat{b}| > \eta + ts_0\zeta\,\hat{a}\cdot\hat{b}$ unless $\hat{b} = \pm\hat{a}$ (and $\eta \pm ts_0\zeta \geq 0$), which is excluded when E_0 and F_0 do not commute, so no state attains both S_1 and S_2. (ii) The statistics depend only on $z = \hat{a}\cdot\vec{r} \in [-1, 1]$. The largest outcome probabilities of the two measurements are $A(z) = \frac{1}{2}(1 + |\mu + \eta z|)$ and $B(z) = \frac{1}{2}(1 + |\nu + \zeta z|)$, so $L_1 = \max\{A(z), B(z)\}$ and $L_2 = A(z) + B(z)$, because $A + B \geq 1$. Both functions are convex, hence their maxima occur at endpoints, giving $S_1 = \max_\varepsilon \max\{A_\varepsilon, B_\varepsilon\}$ and $S_2 = \max_\varepsilon(A_\varepsilon + B_\varepsilon)$. Moreover, if $L_1 = S_1$, the smallest outcome probability is $1 - S_1$, so $L_3 = 1 + S_1 = S_3$ automatically. Thus the vector is attained iff one endpoint saturates both S_1 and S_2. (iii) Here $\kappa = 0.95$ is reached only by the second outcome of N at $\vec{r} = -\hat{a}$, and $\Lambda = 1.55$ only by $(s, t) = (1, -1)$ at $\vec{r} = \hat{a}$. □

Bias and unequal sharpness arise naturally in photon counting. If the two outputs of a polarizing beam splitter are monitored by detectors of efficiencies η_H and η_V, and no-click events are assigned to either outcome at random, the effect of outcome H has the form (6) with $\hat{a} = \hat{z}$, $\mu = (\eta_H - \eta_V)/2$ and $\eta = (\eta_H + \eta_V)/2$. A mismatch of detection efficiencies, a known concern in quantum key distribution [63], thus appears as a bias. A single click/no-click detector behind a polarizer, with no-click events counted as the second outcome, gives $\mu = \eta_d - 1$ and $\eta = \eta_d$, so that $|\mu| + \eta = 1$.

Uncertainty versus incompatibility.—For unbiased POVMs the vectors $\eta\hat{a} \pm \zeta\hat{b}$ that determine Λ also decide joint measurability: the pair is jointly measurable if and only if $|\eta\hat{a} + \zeta\hat{b}| + |\eta\hat{a} - \zeta\hat{b}| \leq 2$ [13, 62]; for biased POVMs the criterion is more involved and no longer depends on these vectors alone [64–66]. The two notions are nevertheless distinct. For orthogonal axes and $\eta = \zeta \leq 1/\sqrt{2}$ the measurements are jointly measurable, yet $S_2 = 1 + \eta/\sqrt{2}$ stays below the value $1 + \eta$ that would be reached if both outcome distributions could be sharpened at once, and $\omega_\oplus(M, N)$ is not attained (Proposition 3). Attainment requires commuting effects, whether or not the pair is jointly measurable, and with bias even commuting effects need not attain the bound: in the example of Proposition 3(iii) the two readouts of the same axis favor opposite ends of it, a trade-off of classical origin. Majorization uncertainty relations therefore quantify preparation uncertainty, which persists in the absence of measurement incompatibility.

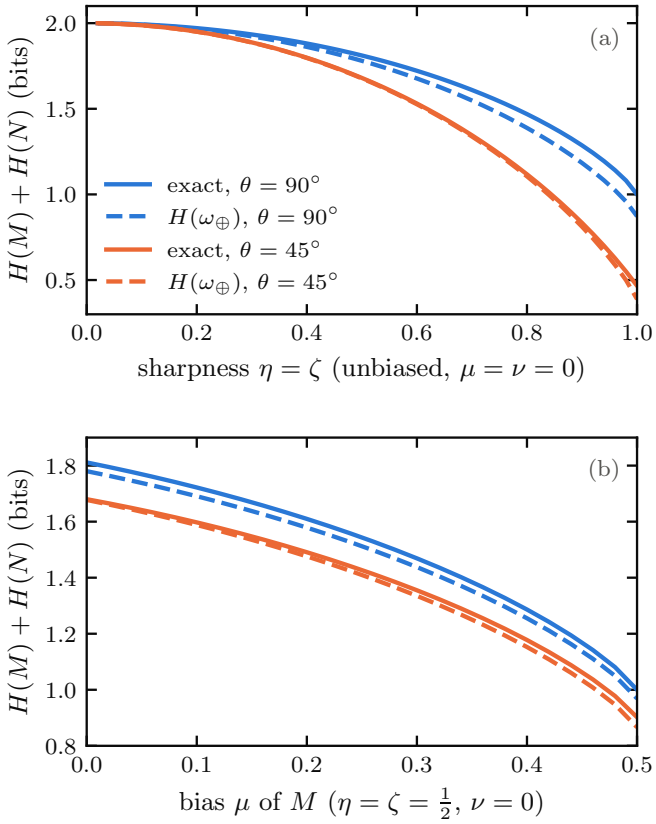


FIG. 5. Exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 4 (dashed) for Bloch angles 90° and 45°. (a) Unbiased measurements of equal sharpness ($\mu = \nu = 0$, $\eta = \zeta$). (b) M with sharpness $\eta = \frac{1}{2}$ and bias μ; N unbiased with $\zeta = \frac{1}{2}$.

As a corollary, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ with $\omega = \omega_\oplus(M, N)$ for all two-outcome qubit POVMs; by Proposition 3 and Theorem 3 this bound is strict whenever E_0 and F_0 do not commute. Figure 5 shows that it tracks the exact minimum as a function of sharpness, becoming exact in the trivial limit $\eta \to 0$, and of bias, where the gap grows with the bias at 45° but not at 90°.

Imperfect detectors and uncertainty-based tests.—Experiments that test uncertainty relations, or use them to certify entanglement or steering, usually compare their data with the bound for ideal measurements. With the

detector models above, Theorem 4 tells exactly when this is justified.

Proposition 4. *Let M, N be two-outcome qubit POVMs with axes $\hat{a}, \hat{b}$, and let M°, N° be the projective measurements along the same axes, with statistics p°, q° and overlap $c = \frac{1}{2}(1 + |\hat{a}\cdot\hat{b}|) < 1$. (i) If $\mu = \nu = 0$, then $p(\rho)\oplus q(\rho) \prec p^\circ(\rho)\oplus q^\circ(\rho)$ for every ρ; hence every lower bound on $\Phi(p\oplus q)$ or on $\Phi(p) + \Psi(q)$, with Φ, Ψ Schur-concave, that holds for M°, N° also holds for M, N. (ii) In general, the ideal relation $p(\rho)\oplus q(\rho) \prec \omega_\oplus$ holds for all ρ if and only if $\Lambda \le 2\sqrt{c}$. If $\Lambda > 2\sqrt{c}$, some state violates the Landau–Pollak relation $p_i + q_j \le 1 + \sqrt{c}$ of the ideal measurements. (iii) If M and N are read out by the same detectors, $\mu = \nu$ and $\eta = \zeta$, with $\hat{a}\cdot\hat{b} \ge 0$, the condition reads $|\mu| \le \sqrt{c}\,(1-\eta)$. Click/no-click detection violates it for every $\eta_d < 1$.*

Proof. (i) For $\mu = 0$, $p = \eta p^\circ + (1-\eta)u$ with $u = (\frac{1}{2}, \frac{1}{2})$, i.e., $p = Dp^\circ$ for the doubly stochastic matrix $D = \frac{1}{2}\left(\begin{smallmatrix} 1+\eta & 1-\eta \\ 1-\eta & 1+\eta \end{smallmatrix}\right)$; likewise $q = D'q^\circ$. Hence $p \prec p^\circ$, $q \prec q^\circ$ and, since $D\oplus D'$ is doubly stochastic, $p\oplus q \prec p^\circ\oplus q^\circ$ [28]; Schur-concavity gives the claim. (ii) By Theorems 1 and 4, $\omega_\oplus$ and $\omega_\oplus(M,N)$ are lattice suprema. If $\omega_\oplus(M,N) \prec \omega_\oplus$, transitivity gives $p\oplus q \prec \omega_\oplus$ for all states; conversely, if $\omega_\oplus$ bounds all $p\oplus q$, then $\omega_\oplus(M,N) = \sup \prec \omega_\oplus$. Since $S_1 = \frac{1}{2}(1+\kappa) \le 1$ and $S_3 = \frac{1}{2}(3+\kappa) \le 2$ always, the relation holds iff $S_2 = 1 + \frac{1}{2}\Lambda \le 1 + \sqrt{c}$. If $\Lambda > 2\sqrt{c}$, the state that attains S_2 gives $p_i + q_j = 1 + \frac{1}{2}\Lambda > 1 + \sqrt{c}$ for some i, j. (iii) For $\mu = \nu$, $\eta = \zeta$ and $\hat{a}\cdot\hat{b} \ge 0$, the terms of Λ are $2|\mu| + \eta|\hat{a} + \hat{b}| = 2|\mu| + 2\eta\sqrt{c}$ for $s = t$ and $\eta|\hat{a} - \hat{b}| \le 2\eta\sqrt{c}$ for $s = -t$, so $\Lambda = 2|\mu| + 2\eta\sqrt{c}$. For click/no-click detection $|\mu| = 1 - \eta_d$, and the condition becomes $(1-\eta_d)(1-\sqrt{c}) \le 0$, which fails for $\eta_d < 1$ and $c < 1$. □

Noise only blurs the statistics towards the uniform distribution, but bias shifts the image of $\mathcal{E}_\theta$, and the shifted ellipse can leave the ideal one. Whether a particular entropic bound is violated depends on the measure: the Shannon minimum is more forgiving than the Landau–Pollak sum, but not immune (Fig. 6). For orthogonal axes and random no-click assignment, an efficiency $\eta_H = 0.99$ violates the ideal relation for $\eta_V < 0.942$ and the ideal Shannon bound of one bit for $\eta_V < 0.578$; for $\eta_H = 0.95$ the thresholds are 0.708 and 0.211, and for $\eta_H = 0.9$ the Shannon bound is never violated, although the ideal relation still is for $\eta_V < 0.417$. Click/no-click detectors lower the Shannon minimum below its ideal value at all efficiencies and angles we examined; for orthogonal axes this holds for every $\eta_d < 1$, since the eigenstate of the dark outcome of M gives $H(M) + H(N) = h(\eta_d/2) < 1$. In all cases $H(\omega_\oplus(M,N))$ remains a valid bound.

This matters for steering tests [10–12], in which Bob trusts his measurements and Alice's are uncharacterized.

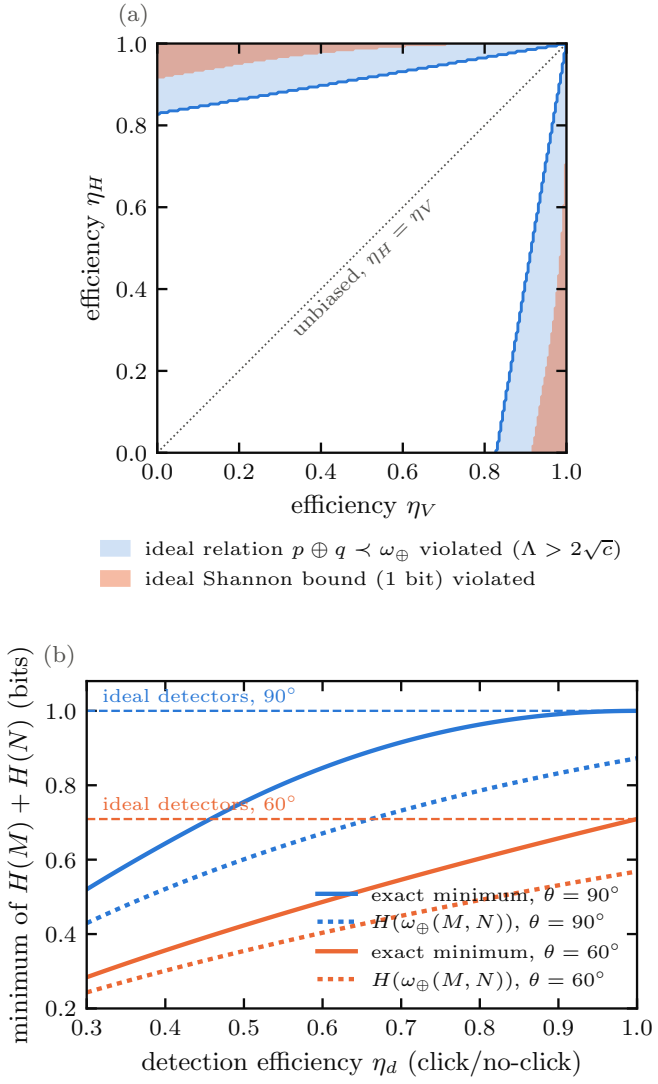


FIG. 6. (a) Polarizing beam splitter with detectors of efficiencies (η_H, η_V) and random assignment of no-click events, for measurements along orthogonal axes read out by the same detectors. Blue: the ideal relation $p\oplus q \prec \omega_\oplus$ fails ($\Lambda > 2\sqrt{c}$, Proposition 4); orange: the ideal Shannon bound of one bit fails. Unbiased detectors ($\eta_H = \eta_V$, dotted) never violate ideal bounds. (b) Click/no-click detection with efficiency η_d: exact minimum of $H(M) + H(N)$ (solid) and the bound $H(\omega_\oplus(M,N))$ of Theorem 4 (dotted) for Bloch angles 90° and 60°; dashed lines mark the minimum for ideal detectors.

Losses and measurement-setting errors have been addressed in other steering criteria [67, 68]; here we use the optimal majorization bound of the POVM actually performed.

Corollary 3. *If Alice's measurements A_1, A_2 and Bob's two-outcome measurements M, N admit a local-hidden-*

state model, then

$$H(M|A_1)+H(N|A_2) \;\geq\; \min_{\rho}\big[H(M)+H(N)\big] \;\geq\; H\big(\omega_{\oplus}(M,N)\big). \quad (27)$$

The minimum for the ideal measurements M°, N° may replace the right-hand side if M, N are unbiased, but not in general: for click/no-click detection along orthogonal axes, a product state with Bob in the dark eigenstate of M gives $H(M|A_1)+H(N|A_2) = h(\eta_d/2)$, below the ideal value of one bit for every $\eta_d < 1$.

Proof. In a local-hidden-state model [12], $p(a,b|x,y) = \sum_\lambda p(\lambda)\, p(a|x,\lambda)\, \mathrm{tr}(\rho_\lambda K_b^y)$, where K_b^y are the effects of Bob's measurement $y \in \{M,N\}$. Conditioning on λ cannot increase entropy, and given λ Bob's outcome is independent of Alice's, so $H(B_y|A_x) \geq H(B_y|A_x\Lambda) = H(B_y|\Lambda) = \sum_\lambda p(\lambda) H(B_y)_{\rho_\lambda}$. Summing over $(x,y) = (1,M),(2,N)$ gives the first inequality in Eq. (27) [10, 11], and Theorem 4 the second. For the example, $\hat{a} = \hat{z}$, $\hat{b} = \hat{x}$, $E_0 = \eta_d|0\rangle\langle 0|$ and Bob in $|1\rangle$: $H(M) = 0$ and $H(N) = h(\eta_d/2)$, independent of Alice's outcomes. For unbiased measurements, Proposition 4(i) shows that the ideal minimum is a valid right-hand side. □

A closed-form Werner-state example illustrates the difference between the ideal and corrected tests. Let $\rho_w = w|\Phi^+\rangle\langle\Phi^+| + (1-w)\mathbb{1}_4/4$, let Alice use ideal σ_z and σ_x measurements, and let Bob use click/no-click detectors of efficiency η_d along the same axes. Then

$$\mathcal{S}(w,\eta_d) := H(M|A_1)+H(N|A_2) = h\left(\frac{\eta_d(1+w)}{2}\right) + h\left(\frac{\eta_d(1-w)}{2}\right). \quad (28)$$

For the orthogonal measurements, Theorem 4 gives $\omega_{\oplus}(M,N) = (1, t_{\eta_d}, 1-t_{\eta_d}, 0)$ with $t_{\eta_d} = 1-\eta_d+\eta_d/\sqrt{2}$, so the corrected bound is $B(\eta_d) = h(t_{\eta_d})$. For the Bell state, $\mathcal{S}(1,\eta_d) = h(\eta_d)$, which violates $B(\eta_d)$ iff

$$\eta_d > \eta_* := \frac{4+\sqrt{2}}{7} \approx 0.773459. \quad (29)$$

Since $\mathcal{S}(w,\eta_d)$ decreases with w, sufficiently large w gives a violation for $\eta_d > \eta_*$, whereas no Werner state is certified by this particular inequality for $\eta_d \leq \eta_*$. Nonviolation of this sufficient witness does not, by itself, imply a local-hidden-state model.

An uncertainty-based steering or entanglement test with biased trusted detectors must therefore use the bound of the measurements actually performed. Theorem 4 provides it in closed form, for every Schur-concave measure at once, and Proposition 4 tells when the ideal bound can be kept.

VII. BINARY PROJECTIVE MEASUREMENTS IN DIMENSION d

The qubit analysis already contains the general binary projective case.

Theorem 5. *Let $M = \{P, \mathbb{1}-P\}$ and $N = \{Q, \mathbb{1}-Q\}$ be projective measurements on $\mathbb{C}^d$, write $P_1 = P$, $P_2 = \mathbb{1}-P$, $Q_1 = Q$, $Q_2 = \mathbb{1}-Q$, and let $c = \max_{i,j} \|P_iQ_j\|^2$, where $\|\cdot\|$ is the operator norm. Then Theorems 1 and 2 hold verbatim, and $\omega_\oplus$ and $\omega_\otimes$ are attained if and only if $c = 1$.*

Proof. By the eigenvalue formula of Sec. VI, S_k for the direct sum is the largest eigenvalue of a sum of k of the four projectors. Hence $S_1 = 1$, $S_3 = 2$, since any three of them include $P_1 + P_2 = \mathbb{1}$ or $Q_1 + Q_2 = \mathbb{1}$, and $S_2 = \max_{i,j} \lambda_{\max}(P_i+Q_j) = 1+\sqrt{c}$, since $\lambda_{\max}(P+Q) = 1 + \|PQ\|$ for any two projectors [69]. Moreover $c \geq \frac{1}{2}$: if $Q_1 \neq 0$ and ψ is a unit vector in its range, then $\|P_1\psi\|^2 + \|P_2\psi\|^2 = 1$, so $\max_i \|P_iQ_1\|^2 \geq \frac{1}{2}$ (and if $Q_1 = 0$, then $Q_2 = \mathbb{1}$ and $c = 1$). The increments therefore give $\omega_\oplus$, which is ordered because $1 \geq \sqrt{c} \geq 1-\sqrt{c}$, so Lemma 1 yields Theorem 1. For the tensor product, $p_iq_j \leq \left(\frac{p_i+q_j}{2}\right)^2 \leq w$, with equality at a top eigenvector of $P_i + Q_j$ with $\|P_iQ_j\|^2 = c$. Indeed, by Jordan's lemma [69], which is also the standard tool for entropic uncertainty relations of binary measurements [49], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces of dimension one or two that are invariant under P and Q; on a two-dimensional subspace with overlap c, the bisector state of the qubit measurements that P and Q induce there gives $p_i = q_j = (1+\sqrt{c})/2$, and for $c = 1$ a common eigenvector gives $p_i = q_j = 1$. Hence $S_1 = w$, while a state in the range of P gives $S_2 = S_3 = S_4 = 1$; since $c \geq \frac{1}{2}$ implies $w \geq \frac{1}{2}$, the increments $(w, 1-w, 0, 0)$ are ordered, and Theorem 2 follows.

If $c < 1$, a state with $L_1(p \oplus q) = 1$ lies in the range of some P_i (or Q_j); then $q_j = \mathrm{tr}(\rho P_iQ_jP_i) \leq \|P_iQ_j\|^2 \leq c$ for all j, so $L_2(p \oplus q) \leq 1 + c < 1 + \sqrt{c}$ and $\omega_\oplus$ is not attained. Likewise, $L_2(p \otimes q) = 1$ requires two vanishing entries of $p \otimes q$, hence such a state, at which $L_1(p \otimes q) \leq c < w$, so $\omega_\otimes$ is not attained either. If $c = 1$, the ranges of some P_i and Q_j intersect, and a common eigenvector attains $\omega_\oplus = (1,1,0,0)$ and $\omega_\otimes = (1,0,0,0)$. □

In particular, all bounds of Sec. V hold for binary projective measurements in any dimension with $c = \max_{i,j} \|P_iQ_j\|^2$, the min-entropy bound remains tight, and for $c < 1$ the bounds for strictly Schur-concave measures are strict (Theorem 3). The direct-sum part of Theorem 5 follows from the eigenvalue formula [37] and $\lambda_{\max}(P+Q) = 1 + \|PQ\|$ [69]. The tensor-product vector itself is the binary case of the general construction of Friedland *et al.* [33]; here we evaluate its partial sums explicitly, show that it is already ordered, and reduce its attainment to qubit blocks by Jordan's lemma [49, 69]. The attainment conditions in dimension d are new. Equivalently, $c = \frac{1}{2} + \frac{1}{4}\|\{A,B\}\|$ with $A = 2P - \mathbb{1}$ and $B = 2Q - \mathbb{1}$, as shown in any dimension by Kaniewski *et al.* [49]. Indeed, on a two-dimensional block, where P and Q act as qubit projectors with axes $\hat{m}_b$ and $\hat{n}_b$, $\{A,B\} = 2(\hat{m}_b \cdot \hat{n}_b)\mathbb{1}$ and $c_b = \frac{1}{2}(1 + |\hat{m}_b \cdot \hat{n}_b|)$, while $\{A,B\} = \pm 2$ on a one-dimensional block. This relation between the overlap and the effective anticommutator

is closely related to the anticommutator norm formula $\|PQ+QP\| = \|PQ\| + \|PQ\|^2$ [70].

The argument for the direct sum uses only the eigenvalue formula, so it carries over to binary POVMs in any dimension.

Corollary 4. *Let $M = \{E_1, E_2\}$ and $N = \{F_1, F_2\}$ be two-outcome POVMs on $\mathbb{C}^d$, let S_1 be the largest eigenvalue of the four effects, and let $S_2 = \max_{i,j} \lambda_{\max}(E_i + F_j)$. Then*

$$p(\rho) \oplus q(\rho) \;\prec\; \big(S_1,\; S_2 - S_1,\; 1 + S_1 - S_2,\; 1 - S_1\big) \qquad \forall \rho, \tag{30}$$

and this vector is the optimal majorization bound.

Proof. By the eigenvalue formula, S_1 and S_2 are the maximal partial sums (a pair within one measurement sums to 1, which never exceeds the cross terms, since $S_2 \geq S_1 + \frac{1}{2}$ as shown below), and $S_3 = 1 + S_1$, since any three effects include $E_1 + E_2 = \mathbb{1}$ or $F_1 + F_2 = \mathbb{1}$. The increments are non-increasing iff $S_1 + \frac{1}{2} \leq S_2 \leq 2S_1$. The upper inequality holds because $\lambda_{\max}$ is subadditive and $S_1 \geq \frac{1}{2}$. For the lower one, a top eigenvector of the effect that gives S_1 is a state at which the other measurement has an outcome of probability at least $\frac{1}{2}$. Lemma 1 completes the proof. □

Theorem 4 is the qubit case, with $S_1 = \frac{1}{2}(1+\kappa)$ and $S_2 = 1 + \frac{1}{2}\Lambda$, and Theorem 5 the projective case, with $S_1 = 1$ and $S_2 = 1 + \sqrt{c}$. Proposition 4(ii) generalizes in the same way: since $S_1 \leq 1$ and $S_3 = 1 + S_1 \leq 2$ always, two binary POVMs obey the relation $p \oplus q \prec (1, \sqrt{c}, 1 - \sqrt{c}, 0)$ of a reference pair of projective measurements with overlap c iff $S_2 \leq 1 + \sqrt{c}$. For arbitrary POVMs in any dimension, the eigenvalue formula together with Lemma 1 gives an exact numerical procedure, which we use to verify Theorems 4 and 5 and Corollary 4 independently (Appendix A).

VIII. DISCUSSION

For binary measurements the majorization approach to uncertainty can be carried out in closed form. For a qubit the achievable statistics form an affine image of an ellipse, and a single four-component vector, determined by the overlap c, is their least upper bound in the majorization lattice, for the direct sum as well as for the tensor product. For POVMs the direct-sum bound is determined by κ and Λ, and in any dimension by the largest eigenvalues of the effects and of their pairwise sums (Corollary 4). This is the unified perspective that majorization offers: one vector controls every Schur-concave uncertainty measure at once.

The comparison with entropy-specific relations shows what this unification costs. Because the supremum is not the statistics of any state when the measurements do not commute, the induced bound is strict for every strictly Schur-concave measure, and for the tensor-product Rényi bounds it becomes exact only at the extreme orders $\alpha = 0$ and $\alpha = \infty$. Against a generic scalar relation such as MU, on the other hand, the majorization bounds are strong. For two projective qubit measurements, both Shannon bounds depend only on the Bloch angle θ between the two measurement axes: the direct-sum bound is $h(\sqrt{c})$ and MU gives $-\log_2 c$, with $c = \cos^2\frac{\theta}{2}$. The two curves cross exactly once, at $\theta^* \approx 81.3°$ ($c \approx 0.575$), the root of $h(\cos\frac{\theta}{2}) = -2\log_2\cos\frac{\theta}{2}$ (Sec. V and Fig. 3). Nothing singles out this angle geometrically; it is simply where the two bounds exchange roles. For $\theta < \theta^*$, that is, unless the measurements are close to mutually unbiased, the direct-sum Shannon bound is the stronger of the two, and it becomes asymptotically tight in ratio as the measurements approach compatibility, precisely where MU is weakest. Only in the narrow window $\theta^* < \theta \leq 90°$ next to mutually unbiased measurements ($\theta = 90°$), for which MU is tight, does MU give the better bound; for the tensor-product bound the crossover lies at 79.3°. For the Shannon entropy alone, the closed-form bound (19) is at least as strong as both at every angle, and $h(\sqrt{c})$ is never below 78% of the exact minimum, which thus exceeds it by up to 28%; the strength of the majorization bounds lies in their universality. Of the two forms, the direct sum is preferable for $\alpha \leq 1$, and for qubits is numerically observed to remain valid up to $\alpha = 2$, while only the tensor product survives for large α.

Majorization also makes the distinction between detector noise and detector bias sharp. Noise acts as a doubly stochastic map on the statistics and can only increase every measure of uncertainty, whereas bias moves the achievable set. Whether relations derived for ideal detectors survive is decided by one partial sum, S_2, that is, by $\Lambda \leq 2\sqrt{c}$ for qubits (Proposition 4). Uncertainty-based tests of entanglement and steering with trusted detectors must therefore account for detector bias explicitly; the closed-form bound of Theorem 4 does so for all Schur-concave measures simultaneously. Non-attainment is itself testable: experiments have reported Lorenz curves that touch the optimal bound [43], which by Proposition 1 can only happen for individual partial sums, and the distance $\min_\rho \max_k [S_k - L_k(p \oplus q)]$ between the bound and the closest state is a directly measurable quantity. A test with deliberately mismatched detectors, in the coherent-light [43] or photonic [42] settings in which optimal majorization bounds have already been probed, would probe Proposition 4 directly.

Relation to quantum optics.—These results contrast instructively with the quadratures of a light mode. There, coherent and squeezed states are minimum-uncertainty states: for their principal quadratures they saturate both the Heisenberg relation and the entropic relation of Białynicki-Birula and Mycielski [18]; moreover, all pure Gaussian states are equivalent under continuous majorization in phase space and are conjectured to majorize every Wigner-positive state [71]. For non-commuting binary measurements no such state exists (Propositions 1 and 3), and the optimal state depends on the uncer-

tainty measure, in line with the finding of Korzekwa and Lostaglio [72] that no qubit state minimizes all information-theoretic uncertainty measures at once, and with the measure dependence of qubit uncertainty discussed in Ref. [60]. The boundary of $\mathcal{E}_\theta$, traced by the pure states in the plane of the two axes, plays the role of the family of squeezed states, trading predictability of N for that of M. The balanced bisector state, the analogue of a coherent state, minimizes the Shannon sum up to $\bar{\theta} \approx 67°$; for larger angles the minimizer moves towards an eigenstate of either measurement, the analogue of an infinitely squeezed state, and the range in which the balanced state is optimal shrinks for smaller Rényi orders and grows for larger ones (Sec. V and Ref. [48]). The ellipse itself resembles the covariance ellipse of a squeezed vacuum only loosely: it describes the pair of measurements rather than a single state, and although it is a disk for mutually unbiased measurements and flattens as they become compatible, its area $\pi \sin\theta$ shrinks instead of being conserved. A single qubit cannot be squeezed in the usual sense [73], as all its pure states are spin coherent states; the essential difference is algebraic. For quadratures the commutator is a c-number, which makes the Heisenberg bound state independent. For binary observables it is the anticommutator that is, blockwise, a c-number [Eq. (9)], and it fixes $c = \frac{1}{2} + \frac{1}{4}\|\{A, B\}\|$ (Sec. VII). The Robertson bound $\Delta A\, \Delta B \geq |\langle(\hat{m} \times \hat{n}) \cdot \vec{\sigma}\rangle|$, by contrast, vanishes for every state whose Bloch vector lies in the plane of the two axes, which contains all optimal states. Commutator-based relations are therefore uninformative here, and state-independent descriptions are needed: tight variance relations [53] or, for all Schur-concave measures at once, majorization. The information missing from the Robertson bound is carried by the anticommutator term of the Robertson–Schrödinger relation, $\det\Gamma \geq \langle(\hat{m} \times \hat{n}) \cdot \vec{\sigma}\rangle^2$ with Γ from Eq. (9), which every pure qubit state saturates for these Pauli observables; in the plane of the axes it reduces to $\det\Gamma = 0$, the boundary of $\mathcal{E}_\theta$.

The results apply directly to photonic qubits. Restricted to single photons, the Stokes operators of polarization are the Pauli matrices, and two linear polarization analyzers at a relative angle ϑ correspond to $\theta = 2\vartheta$, so that $c = \cos^2\vartheta$ is Malus's law; the crossover with MU and the angle $\bar{\theta}$ correspond to $\vartheta \approx 40.7°$ and $\vartheta \approx 33.5°$, respectively. The optimal direct-sum bound has been tested in this setting with coherent light, the polarization state being reconstructed from measured Stokes parameters [43]. For many photons the Stokes operators form a collective spin that can be squeezed (polarization squeezing [74, 75]); for a single photon, which cannot be squeezed, only the trade-off along the boundary of $\mathcal{E}_\theta$ remains. Imperfect detectors lead to the unsharp and biased measurements of Theorem 4, with the bias set by the mismatch of detection efficiencies (Sec. VI); by Proposition 3(iii), such biases can create a trade-off even between two measurements in the same polarization basis, and by Proposition 4 they can invalidate the ideal-detector bounds.

Several questions remain open. (i) *Tensor bounds for POVMs.* The optimal $\omega_\otimes$ requires maximizing sums of products of affine functions over $\mathcal{E}_\theta$, and we have not found it for any noisy measurements. For unbiased measurements of equal sharpness the AM–GM step of Theorem 2 still gives S_1, but S_3 is no longer attained at the states that saturate the projective bounds: numerically, for orthogonal axes and $\eta = \zeta = 0.8$ it is attained neither by an eigenstate nor by a bisector state, and we have not derived it in closed form. For biased or unequally sharp measurements the AM–GM step itself is no longer tight, and the least concave majorant in Lemma 1 may become non-trivial. (ii) *Binary POVMs in dimension d.* Corollary 4 gives the optimal direct-sum bound in terms of $\lambda_{\max}(E_i + F_j)$. A closed form for this eigenvalue, analogous to $1 + \|P_i Q_j\|$ for projectors, the conditions under which the bound is attained, and the tensor-product bound remain open. (iii) *More measurements.* For qubit bases the eigenvalue formula gives $S_k = \frac{k}{2} + \frac{1}{2}\max_J \left|\sum_{j\in J} \vec{n}_j\right|$, where J runs over sets of k outcomes and $\vec{n}_j$ is the Bloch vector of the projector of outcome j, and Li and Qiao obtained the optimal bound for σ_x, σ_y and σ_z [40]. For random triples of bases we found the increments of S_k always ordered, so that no flattening was needed; whether this holds in general, and what replaces it for POVMs and for the tensor product, is open. (iv) *Mixed orders.* For qubits, conditional majorization bounds yield optimal relations for any pair of orders [41]. Whether state-independent relations of the form (3) can reproduce mixed-order relations such as $H_\infty(p) + H_{1/2}(q) \geq -\log_2 c$ remains open. (v) *Binary measurements on light modes.* Which experimentally accessible binary measurements on a light mode have $c < 1$, and what do the bounds of Sec. VII imply for them? In infinite dimension $c = \sup_{i,j} \|P_i Q_j\|^2$ can equal 1 without a common eigenvector. For photon-number parity and a suitably displaced parity, whose expectation values sample the Wigner function [76], increasingly squeezed vacua, which contain only even photon numbers, approach simultaneous eigenstates of both when the displacement is along the anti-squeezed quadrature, so no non-trivial state-independent relation exists. Tests for coherent states, $\{|\nu\rangle\langle\nu|, \mathbb{1} - |\nu\rangle\langle\nu|\}$, realized by a displacement followed by on/off detection with no-click probability $\pi Q(\nu) = \langle\nu|\rho|\nu\rangle$, with Q the Husimi function, also have $c = 1$, because all states orthogonal to both coherent states are common eigenvectors. Individual partial sums remain informative, however: $\pi[Q(\nu_1) + Q(\nu_2)] \leq 1 + e^{-|\nu_1 - \nu_2|^2/2}$, with equality for the superposition $|\nu_1\rangle + e^{-i\varphi}|\nu_2\rangle$ with $\varphi = \arg\langle\nu_1|\nu_2\rangle$, an optical cat state [77, 78] (the even cat state for $\nu_2 = -\nu_1$), which thus plays the role of the bisector state. Non-trivial relations require $\|\{A, B\}\| < 2$, as for the pseudospin operators of Ref. [79], or for polarization measurements on single photons.

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Appendix A: Numerical methods

All analytical results were also checked numerically. We evaluated Lorenz curves, the lattice supremum via least concave majorants [56], the analytic bounds of Theorems 1, 2 and 4 and Corollaries 1–2, the eigenvalue characterization $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j\in J} K_j)$, and numerical minima of Rényi and Tsallis entropy sums.

The closed forms were compared with (a) lattice suprema over 1.8×10^5 grid states, for 40 random pairs of qubit bases (Theorems 1 and 2) and 60 random pairs of qubit POVMs (Theorem 4), with maximal deviation $< 10^{-4}$, and (b) the eigenvalue characterization for 200 random qubit POVM pairs and for 150 random pairs of projectors of rank $d/2$ in dimensions $d = 2, 4, 6$, all with $c < 1$ (deviation $< 10^{-14}$). Pairs with a common eigenvector, which always occur unless both ranks equal $d/2$, have $c = 1$ and were checked separately; the anticommutator form of c was checked for 300 random pairs of projectors of arbitrary rank in dimensions $d \le 8$, the tensor-product part of Theorem 5 for 50 random pairs in $d = 4, 6$, and Corollary 4 for 300 random pairs of binary POVMs in $d = 2, \ldots, 6$. We also checked Propositions 1–3, Corollaries 1–2, Theorem 3, the critical angle $\bar{\theta}$, the motion of the Shannon minimizer beyond $\bar{\theta}$, the range of Rényi orders in which the bisector or the eigenstates are optimal, the other values quoted in Sec. V, the covariance form of Lemma 2, the jointly measurable example, the detector model and the behavior of the gap in Fig. 5(b) of Sec. VI, the Robertson–Schrödinger statement of Sec. VIII, and the numerical statements in open questions (i) and (iii), the latter for 300 random triples of qubit bases. The commuting attainment criterion of Proposition 3 was checked for 2000 random POVM pairs against direct evaluations of the endpoint statistics. Proposition 4 was checked for 400 random biased POVM pairs (the criterion $\Lambda \le 2\sqrt{c}$ against the largest Landau–Pollak sum over 8×10^3 states), for 100 random unbiased pairs (state-by-state majorization) and for the detector models of Fig. 6. The Werner-state conditional entropy formula and the threshold η_* were checked against Born-rule matrices and the analytic POVM bound. We also checked the validity of the direct-sum Rényi form up to $\alpha = 2$, the order $\alpha_0(c)$, the tensor-product ratio and the fixed-spectrum bound of Sec. VI. The statements about light modes in Sec. VIII were checked in a Fock space of dimension 600.

The numerical entropy minima are obtained by a dense grid over the great circle in $\mathrm{span}\{\hat{m}, \hat{n}\}$ refined by Nelder–Mead. For Rényi and Tsallis sums this reduction is justified by the radial argument of Sec. III, which applies equally to unbiased unsharp measurements. For biased measurements we minimize only the Shannon sum, which is concave in (x, y) and is therefore minimized on the boundary of $\mathcal{E}_\theta$ as well. To check that the eigenstates are optimal for $\alpha \le \frac{1}{2}$, where the Rényi sums have a cusp at the eigenstates, we parametrize the circle by an angle φ and write the outcome probabilities as $\cos^2(\varphi/2)$ and $\sin^2(\varphi/2)$, which avoids the loss of precision in $1 - \cos\varphi$.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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